Learning Evidence

Exercise 2.1.1 Find the sum of matrices A and B :

1. $A = \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 \\ -7 & 3 \end{bmatrix}$.

$$A+B = \begin{bmatrix} 1-1 & 4+5 \\ 2-7 & -1+3 \end{bmatrix} = \begin{bmatrix} 0 & 9 \\ -5 & 2 \end{bmatrix}$$

2. $A = \begin{bmatrix} 1 & 7 \\ 9 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 0 & -3 \end{bmatrix}$

$$A+B = \begin{bmatrix} 1+2 & 7-1 \\ 9+0 & -2-3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & -5 \end{bmatrix}$$

Exercise 2.1.2 Let $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 \\ 0 & 3 \end{bmatrix}$, find 4A and 7B :

$$4A = \begin{bmatrix} 1\times4 & 0\times4 \\ 2\times4 & -1\times4 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 8 & -4 \end{bmatrix}$$

$$7B = \begin{bmatrix} -1\times7 & 1\times7 \\ 0\times7 & 3\times7 \end{bmatrix} = \begin{bmatrix} -7 & 7 \\ 0 & 21 \end{bmatrix}$$

this is just really simple so far,
associative laws & distributive laws apply.

Matrix Multiplication

$$A = 3\left\{\underbrace{\begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}}_{3}\right., \quad B = 3\left\{\underbrace{\begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}}_{2}\right.$$

$AB \Rightarrow$ 3×3 $\qquad$ 3×2

because these two are identical, we can multiply

this means this exists

$$\begin{array}{cc} & \overset{3}{\begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}} \\ 3 \rightarrow \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} & \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix} \end{array}$$

$2 \cdot 8 + 3 \cdot 7 + 5 \cdot 6 = 67$

$2 \cdot 9 + 3 \cdot 2 + 5 \cdot 1 = 29$

$1 \cdot 8 + 4 \cdot 7 + 7 \cdot 6 = 78$

$1 \cdot 9 + 4 \cdot 2 + 7 \cdot 1 = 24$

$0 \cdot 8 + 1 \cdot 7 + 8 \cdot 6 = 55$

$0 \cdot 9 + 1 \cdot 2 + 8 \cdot 1 = 10$

$$\therefore AB = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$$

$BA \Rightarrow$

$$\begin{array}{cc} & \overset{3}{\begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}} \\ \overset{2}{\rightarrow} \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix} & \end{array}$$

3,2 – not (3,3)

$\therefore$ this cannot exist.

for ipynb sagemath.

Extra Exercises + SageMath Code

- Addition: $\begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$ → `matrix([[1,3],[0,-2]])+matrix([[2,-1],[0,1]])`
  - Output/Ans:
    ```
    [ 3  2]
    [ 0 -1]
    ```
- Scalar Multiplication: $3 * \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$ → `3*matrix([[1,3],[0,2]])`
  - Output/Answer:
    ```
    [3 9]
    [0 6]
    ```
- Matrix Multiplication: $\begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} * \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$ → `matrix([[1,3],[0,-2]])*matrix([[2,-1],[0,1]])`
  - Output/Answer:
    ```
    [ 2  2]
    [ 0 -2]
    ```

$$A = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

AB →

$$\begin{bmatrix} 2 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & -3 & 1 \\ -3 & 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} -7 \\ 0 \\ -6 \end{bmatrix}$$

$2 \cdot 2 + -1 \cdot 1 + 3 \cdot 0 + 5 \cdot -2 = -7$

$0 \cdot 2 + 2 \cdot 1 + -3 \cdot 0 + 1 \cdot -2 = 0$

$-3 \cdot 2 + 4 \cdot 1 + 1 \cdot 0 + 2 \cdot -2 = -6$

$$\therefore AB = \begin{bmatrix} 2 \\ 1 \\ 0 \\ -2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \text{ and } B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix},$$

AB →

$$\begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$$

$2 \cdot 8 + 3 \cdot 7 + 5 \cdot 6 = 67$

$2 \cdot 9 + 3 \cdot 2 + 5 \cdot 1 = 29$

$1 \cdot 8 + 4 \cdot 7 + 7 \cdot 6 = 78$

$1 \cdot 9 + 4 \cdot 2 \cdot 7 \cdot 1 = 24$

$0 \cdot 8 + 1 \cdot 7 + 8 \cdot 6 = 55$

$0 \cdot 9 + 1 \cdot 2 + 8 \cdot 1 = 10$

$$\therefore AB = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix},$$

$AB \Rightarrow$

$$\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 0 & -2 \end{bmatrix}$$

$1 \cdot 2 + 3 \cdot 0 = 2$

$1 \cdot -1 + 3 \cdot 1 = 2$

$2 \cdot 0 - 2 \cdot 0 = 0$

$0 \cdot -1 + -2 \cdot 1 = -2$

$\therefore AB = \begin{bmatrix} 2 & 2 \\ 0 & -2 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 9 & 7 \\ -1 & 0 & 2 \end{bmatrix}$$

$AB \Rightarrow$

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 9 & 7 \\ -1 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & 4 \end{bmatrix}\begin{bmatrix} -1 & -6 & -2 \\ 0 & 6 & 10 \end{bmatrix}$$

$1 \cdot 2 + -1 \cdot 1 + 2 \cdot -1 = -1$

$3 - 9 + 0 = -6$

$1 - 7 + 4 = -2$

$4 + 0 - 4 = 0$

$6 + 0 + 0 = 6$

$2 + 0 + 8 = 10$

$AB = \begin{bmatrix} -1 & -6 & -2 \\ 0 & 6 & 10 \end{bmatrix}$

determinants

$A = \begin{bmatrix} 2 & 1 \\ 3 & -4 \end{bmatrix}$ Find det(A) $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$\det(A) = ad - bc$

$= 2(-4) - 1 \cdot 3 = -11$ //

Find the inverse of $\begin{bmatrix} 1 & 1 \\ 6 & 1 \end{bmatrix}$ (mod 26)

inverse only exists if $\gcd(\det(A), n) = 1$ ← a rule

$\det(A) = 1 \cdot 1 - 6 \cdot 1 = -5$, $\gcd(-5, 26) = 1$ ∴ inverse exists

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \Rightarrow \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \Rightarrow -\frac{1}{5} \begin{bmatrix} 1 & -1 \\ -6 & 1 \end{bmatrix} \pmod{26}$$

$-5 \equiv 21 \pmod{26} \Rightarrow 21^{-1} \bmod 26 \Rightarrow 26 = 1 \times 21 + 5$

$1 = \underline{5} \times 21 - 4 \times 26$

$$5 \begin{bmatrix} 1 & -1 \\ -6 & 1 \end{bmatrix} = \begin{bmatrix} 5 & -5 \\ -35 & 5 \end{bmatrix} \equiv \begin{bmatrix} 5 & 21 \\ 22 & 5 \end{bmatrix}$$ //

Cofactor

$C_{ij}(A) = (-1)^{i+j} \det(A_{ij})$ ⇐ i = row, j = column

$A = \begin{bmatrix} 2 & 1 & 3 \\ 5 & -7 & 1 \\ 3 & 0 & 6 \end{bmatrix}$, $C_{12}(A) = (-1)^{1+2} \det(A_{12})$

$= (-1)^3 \det\left(\begin{bmatrix} 5 & 1 \\ 3 & -6 \end{bmatrix}\right)$

$A_{12} = 1$, $A_{32} = 0$

as i (row) = 1 & j (col) = 2, ignore they exist to get

$= (-1)(5 \cdot -6 - 1 \cdot 3)$

$= -(-33)$

$= 33$ //

Adjugates

$$\text{adj } A = \left[C_{ij}(A) \right]^T$$ ← this is called the transpose

eg ⇒ $A = \begin{bmatrix} 2 & 1 & 3 \\ 5 & -7 & 1 \\ 3 & 0 & -6 \end{bmatrix}$

$C_{11}(A) = (-1)^{1+1} \det \left(\begin{bmatrix} -7 & 1 \\ 0 & -6 \end{bmatrix} \right) = 42$

$C_{12}(A) = (-1)^{1+2} \det \left(\begin{bmatrix} 1 & 3 \\ 0 & -6 \end{bmatrix} \right) = 6$

$C_{13} = 22, \; C_{21} = 33, \; C_{22} = -21, \; C_{23} = 13, \; C_{31} \dots$

do this to find cofactors of everything

then, fill in everything ⇒

$$C(A) = \begin{bmatrix} 42 & 33 & 21 \\ 6 & -21 & 3 \\ 22 & 13 & -19 \end{bmatrix}$$

then, to get the transpose, swap them diagonally ⇒ across this axis

$$\therefore \left[C(A) \right]^T = \begin{bmatrix} 42 & 6 & 22 \\ 33 & -21 & 13 \\ 21 & 3 & -19 \end{bmatrix} = \text{adj } A$$

Determinants of a 3×3 Matrix

eg ⇒ $A = \begin{bmatrix} 2 & 1 & 3 \\ 5 & -7 & 1 \\ 3 & 0 & -6 \end{bmatrix}$

expand along a row

expanding on third row because there was a zero

$$\det(A) = 3 \times C_{31} + 0 \times C_{32} + (-6) \times C_{33}$$

$$= 3 \times (-1)^{3+1} \det \begin{pmatrix} 1 & 3 \\ -7 & 1 \end{pmatrix} + 0 - 6 \times (-1)^{3+3} \det \begin{pmatrix} 2 & 1 \\ 5 & -7 \end{pmatrix}$$

$$= 3 \times 1 (1 - -21) - 6(-14 - 5)$$

$$= 180$$

Inverse of a 3×3 Matrix

$$A^{-1} = \frac{1}{\det A} \times \text{adj } A.$$

calculate both of these

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 5 & -7 & 1 \\ 3 & 0 & -6 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{180}\begin{bmatrix} 42 & 6 & 22 \\ 33 & -21 & 13 \\ 21 & 3 & -19 \end{bmatrix} = \begin{bmatrix} 7/30 & 1/30 & 4/90 \\ 11/60 & -7/60 & 13/180 \\ 7/60 & 1/60 & -19/180 \end{bmatrix}$$

* Exercise 23.1 $\Rightarrow$

① $A = \begin{bmatrix} 5 & -3 \\ 7 & 4 \end{bmatrix} \Rightarrow \det(A) = 5\times 4 - (-3)\times 7 = 41$

② $A = \begin{bmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{bmatrix} \Rightarrow \det(A) = 1\times C_{31} + 3\times C_{32} + 0\times C_{33}$

$$= 1\times(-1)^{3+1}\times\det\begin{pmatrix} 7 & 1 \\ 4 & -1 \end{pmatrix} + 3\times(-1)^{3+2}\times\det\begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$$

$$= 1\times 1\times(-7-4) + 3\times -1\times(-2+4)$$

$$= -2$$

$\text{adj}\, A = \left[C_{ij}(A)\right]^T$

$C_{ij}(A) = (-1)^{i+j}\det(A_{ij})$

* Exercise 2.4.1

① $A = \begin{bmatrix} 5 & -3 \\ 7 & 4 \end{bmatrix} \Rightarrow$

$C_{11}(A) = (-1)^2\times 4 = 4$, $C_{12}(A) = (-1)^3\times 7 = -7$, $C_{21}(A) = -1\times -3 = 3$

$C_{22}(A) = 1\times 5 = 5$

$$C(A) = \begin{bmatrix} 4 & -7 \\ 3 & 5 \end{bmatrix} \Rightarrow \text{adj}\, A = \begin{bmatrix} 4 & 3 \\ -7 & 5 \end{bmatrix}$$

② $B = \begin{bmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{bmatrix}$

$C_{11}(B) = (-1)^2\times\det\begin{pmatrix} 4 & -1 \\ 3 & 0 \end{pmatrix} = 1\times(0+3) = 3$

$C_{12}(B) = (-1)^3\det\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} = -1\times(0+1) = -1$

$C_{13}(B) = (-1)^4\det\begin{pmatrix} 1 & 4 \\ 1 & 3 \end{pmatrix} = 1\times(3-4) = -1$

$C_{21}(B) = (-1)^3\det\begin{pmatrix} 7 & 1 \\ 3 & 0 \end{pmatrix} = -1\times(0-3) = +3$

$$C_{22}(B) = (-1)^4 \det\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = 1 \times (0-1) = -1$$

$$C_{23}(B) = (-1)^5 \det\begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix} = -1 \times (6-7) = 1$$

$$C_{31}(B) = (-1)^4 \det\begin{pmatrix} 7 & 1 \\ 4 & -1 \end{pmatrix} = 1 \times (-7-4) = -11$$

$$C_{32}(B) = (-1)^5 \det\begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} = -1 \times (-2-1) = 3$$

$$C_{33}(B) = (-1)^6 \det\begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix} = +1 \times (8-7) = 1$$

$$\text{adj } A = \begin{bmatrix} 3 & -1 & -1 \\ 3 & -1 & 1 \\ -11 & 3 & 1 \end{bmatrix}^T = \begin{bmatrix} 3 & 3 & -11 \\ -1 & -1 & 3 \\ -1 & 1 & 1 \end{bmatrix} //$$

Inverses

① $A = \begin{bmatrix} 5 & -3 \\ 7 & 4 \end{bmatrix}$, $\quad A^{-1} = \dfrac{1}{\det A} \times \text{adj } A$

$$A^{-1} = \frac{1}{20+21} \times \begin{bmatrix} 4 & 3 \\ -7 & 5 \end{bmatrix} = \frac{1}{41}\begin{bmatrix} 4 & 3 \\ -7 & 5 \end{bmatrix}$$

② $B = \begin{bmatrix} 2 & 7 & 1 \\ 1 & 4 & -1 \\ 1 & 3 & 0 \end{bmatrix}$

$$\det(B) = 1 \times C_{31} + 3 \times C_{32} + 0 \times C_{33}$$
$$= 1 \times -11 + 3 \times 3 + 0$$
$$= -2$$

$$B^{-1} = \frac{1}{-2}\begin{bmatrix} 3 & 3 & -11 \\ -1 & -1 & 3 \\ -1 & 1 & 1 \end{bmatrix}$$

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Matrix Addition in Modular Arithmetic

Exercise 2.5.1

① $$A = \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 5 \\ -7 & 3 \end{bmatrix} \qquad (A+B) \bmod 3 = ?$$

$$A+B = \begin{bmatrix} 1-1 & 4+5 \\ 2-7 & -1+3 \end{bmatrix} = \begin{bmatrix} 0 & 9 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} \pmod 3$$

② $$A = \begin{bmatrix} 1 & 7 \\ 9 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 \\ 0 & -3 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 1+2 & -1+7 \\ 9+0 & -2-3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & -5 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 3 \end{bmatrix} \pmod 4$$

Exercise 2.5.2

① $$A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix} \Rightarrow 4A = \begin{bmatrix} 8 & 12 & 20 \\ 4 & 16 & 28 \\ 0 & 4 & 32 \end{bmatrix} = \begin{bmatrix} 8 & 1 & 9 \\ 4 & 5 & 6 \\ 0 & 4 & 10 \end{bmatrix} \pmod{11}$$

② $$B = \begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix} \Rightarrow 7B = \begin{bmatrix} 56 & 63 \\ 49 & 14 \\ 42 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 8 \\ 5 & 3 \\ 9 & 7 \end{bmatrix} \pmod{11}$$

③ $$C = \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix} \Rightarrow 5C = \begin{bmatrix} 5 & 15 \\ 0 & -10 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \pmod 3$$

④ $$D = \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} \Rightarrow 8D = \begin{bmatrix} 16 & -8 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \pmod 5$$

Exercise 2.5.3

①

$$\begin{bmatrix} 2 & 3 & 5 \\ 1 & 4 & 7 \\ 0 & 1 & 8 \end{bmatrix}\begin{bmatrix} 8 & 9 \\ 7 & 2 \\ 6 & 1 \end{bmatrix} = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix}$$

$2 \cdot 8 + 3 \cdot 7 + 5 \cdot 6 = 67$

$2 \cdot 9 + 3 \cdot 2 + 5 \cdot 1 = 29$

$1 \cdot 8 + 4 \cdot 7 + 7 \cdot 6 = 78$

$1 \cdot 9 + 4 \cdot 2 + 7 \cdot 1 = 24$

$0 \cdot 8 + 1 \cdot 7 + 6 \cdot 8 = 55$

$0 \cdot 9 + 1 \cdot 2 + 8 \cdot 1 = 10$

$$AB = \begin{bmatrix} 67 & 29 \\ 78 & 24 \\ 55 & 10 \end{bmatrix} = \begin{bmatrix} 15 & 3 \\ 0 & 24 \\ 3 & 10 \end{bmatrix} \pmod{26}$$

②

$$\begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 0 & -2 \end{bmatrix}$$

$1 \cdot 2 + 3 \cdot 0 = 2$

$-1 + 3 = 2$

$6 + 0 = 0$

$0 - 2 = -2$

$$AB = \begin{bmatrix} 2 & 2 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 0 & 9 \end{bmatrix} \pmod{11}$$

in the provided solutions, this is a -2

but isn't it always good practice to write final "mod" answers in all >0 numbers?

· just do "det(A) mod x" and scalar multiply & mod.

- Exercise 2.5.4 ⇒

41 mod 3 = 2

① $A^{-1} = \frac{1}{41}\begin{bmatrix} 4 & 3 \\ -7 & 5 \end{bmatrix} = 2\begin{bmatrix} 4 & 3 \\ -7 & 5 \end{bmatrix} = \begin{bmatrix} 8 & 6 \\ -14 & 10 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \pmod 3$

② B^{-1} ⇒ det(B) mod 5 ⇒ 2 mod 5 = 2

$$B^{-1} = 2\begin{bmatrix} 3 & 3 & -11 \\ -1 & -1 & 3 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 6 & -22 \\ -2 & -2 & 6 \\ -2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 3 & 3 & 1 \\ 3 & 2 & 2 \end{bmatrix} \pmod 5$$